VLSI DESIGN INTERNSHIP – TASK 2

Verilog HDL Codes and Testbench Programs

Submitted By:

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Tools Used:

- Xilinx Vivado
- CircuitVerse

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1. AND Gate

Verilog Code:

```
module and_gate(
input A,
input B,
output Y
);
assign Y = A & B;
endmodule
```

---

2. OR Gate

Verilog Code:

```
module or_gate(
input A,
input B,
output Y
);
assign Y = A | B;
endmodule
```

---

3. NOT Gate

Verilog Code:

```
module not_gate(
input A,
output Y
);
assign Y = ~A;
endmodule
```

---

4. NAND Gate

Verilog Code:

```
module nand_gate(
input A,
input B,
output Y
```

);

assign Y = ~(A & B);

endmodule

---

5. NOR Gate

Verilog Code:

module nor_gate(

input A,

input B,

output Y

);

assign Y = ~(A | B);

endmodule

---

6. XOR Gate

Verilog Code:

module xor_gate(

input A,

input B,

output Y

);

assign Y = A ^ B;

endmodule

---

7. Half Adder

Verilog Code:

```
module half_adder(
input A,
input B,
output SUM,
output CARRY
);
assign SUM = A ^ B;
assign CARRY = A & B;
endmodule
```

Half Adder Testbench:

```
`timescale 1ns / 1ps
module half_adder_tb;
reg A;
reg B;
wire SUM;
wire CARRY;
half_adder uut(
.A(A),
.B(B),
.SUM(SUM),
```

```
.CARRY(CARRY)
);
initial
begin
A=0; B=0;
#10;
A=0; B=1;
#10;
A=1; B=0;
#10;
A=1; B=1;
#10;
$finish;
end
endmodule
```

---

8. Full Adder

Verilog Code:

```
module full_adder(
input A,
input B,
input Cin,
output SUM,
output CARRY
```

```
);
assign SUM = A ^ B ^ Cin;
assign CARRY =
(A & B) |
(B & Cin) |
(A & Cin);
endmodule
```

Full Adder Testbench:

```
`timescale 1ns / 1ps
module full_adder_tb;
reg A;
reg B;
reg Cin;
wire SUM;
wire CARRY;
full_adder uut(
.A(A),
.B(B),
.Cin(Cin),
.SUM(SUM),
.CARRY(CARRY)
);
```

```
initial
begin
A=0; B=0; Cin=0;
#10;
A=0; B=0; Cin=1;
#10;
A=0; B=1; Cin=0;
#10;
A=0; B=1; Cin=1;
#10;
A=1; B=0; Cin=0;
#10;
A=1; B=0; Cin=1;
#10;
A=1; B=1; Cin=0;
#10;
A=1; B=1; Cin=1;
#10;
$finish;
end
endmodule
```

Expected Simulation Results

Half Adder:

A| B| SUM| CARRY

0| 0| 0| 0

0| 1| 1| 0

1| 0| 1| 0

1| 1| 0| 1

Full Adder:

A| B| Cin| SUM| CARRY

0| 0| 0| 0| 0

0| 0| 1| 1| 0

0| 1| 0| 1| 0

0| 1| 1| 0| 1

1| 0| 0| 1| 0

1| 0| 1| 0| 1

1| 1| 0| 0| 1

1| 1| 1| 1| 1

Conclusion:

Successfully implemented AND, OR, NOT, NAND, NOR, XOR, Half Adder, and Full Adder circuits using Verilog HDL. Functional verification was performed using dedicated testbenches and simulation waveforms in Xilinx Vivado.